BASF
We create chemistry

Fourtiva™

Next generation catalyst delivers superior butylenes yield with selectivity over propylene

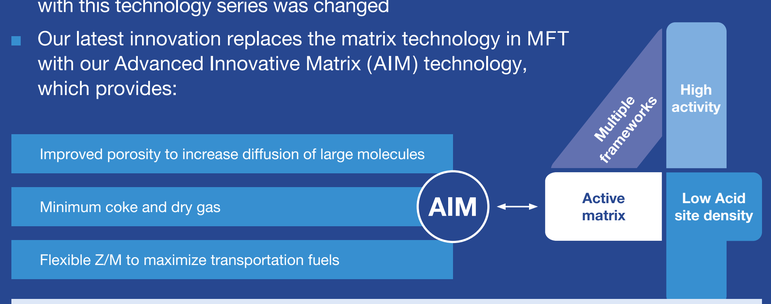
- Fourtiva™ is specifically designed for gasoil to mild resid feedstock
- Optimized to maximize butylene yields and selectivity while improving naphtha octane & LPG olefinicity and minimizing coke and dry gas
- Commercial trials confirmed its superior economic performance helping refiner increase profitability

Benefits of Fourtiva™

- Maximized butylenes yield and selectivity
- Improved naphtha octane & LPG olefinicity
- Minimize coke and dry gas to reduce carbon footprint
- Flexible zeolite to matrix content to maximize transportation fuels

AIM and MFT Technologies Combined

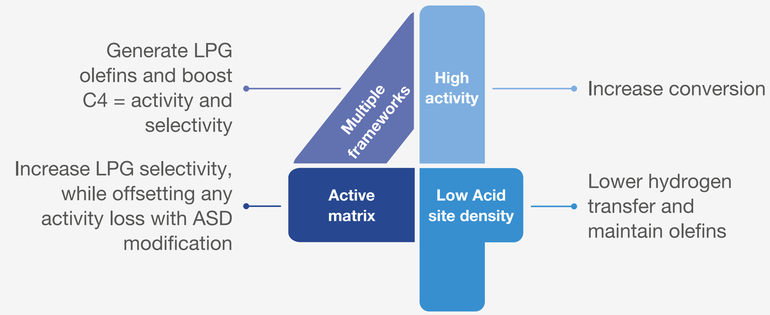
- To improve on existing success of MFT, the matrix design with this technology series was changed
- Our latest innovation replaces the matrix technology in MFT with our Advanced Innovative Matrix (AIM) technology, which provides:



Fourtiva™ incorporates BASF's Advanced Innovative Matrix (AIM) and the Multiple Frameworks Topology (MFT) technology.

Introducing Fourtiva™ Fluidized Catalytic Cracking (FCC) catalyst

to increase high octane gasoline blending feedstock

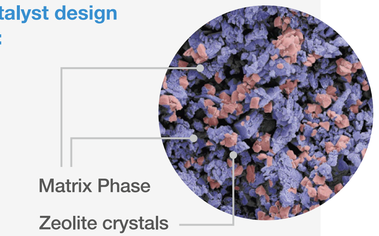


BASF continues to improve matrix technologies

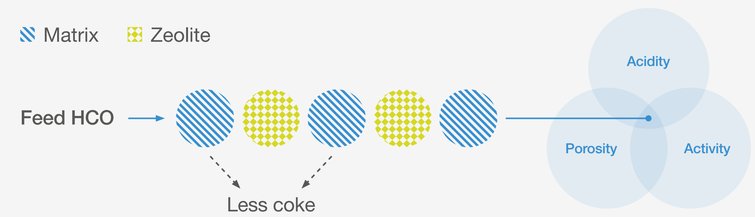
Advanced Innovative Matrix (AIM)

The matrix technology enables catalyst design with improved coke selectivity via:

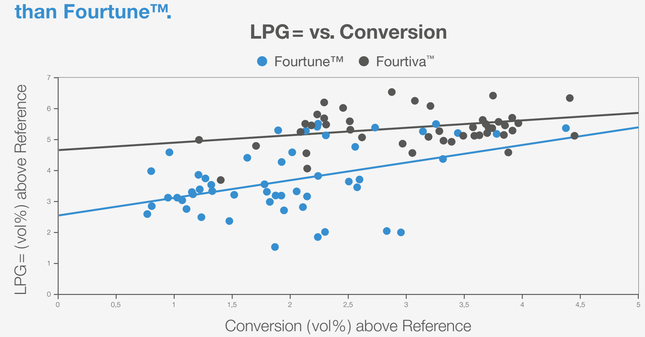
- Optimal matrix acidity
- Optimal matrix activity
- Improved pore size distribution
- Increased matrix/zeolite effectiveness

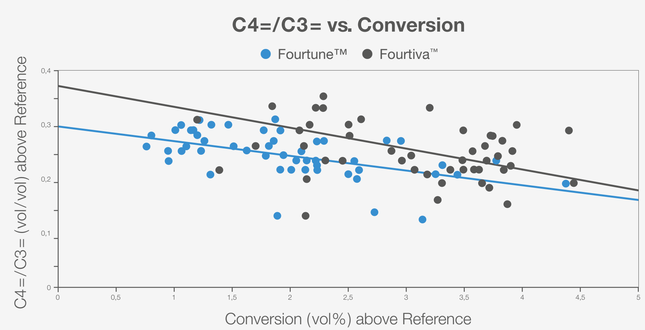


BASF technology for butylene: Multiple Framework Topologies (MFT)

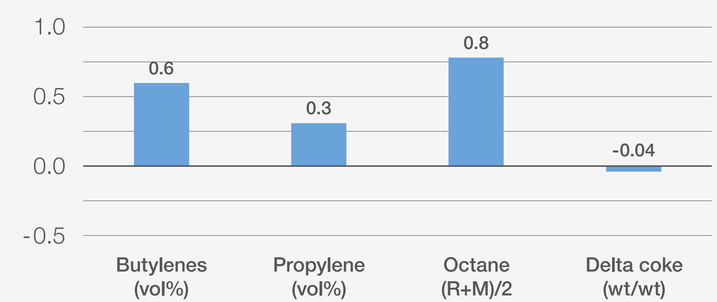


At similar feed rate and olefin additive rates to the unit, the total LPG='s are higher with better selectivity for C4='s for Fourtiva™ than Fourtune™.





Unit Operations Yield Data, Fourtiva™ vs. Fourtune™



- Maximized butylenes yield and selectivity versus other butylene catalysts (Fourte® and Fourtune™)
- Commercial trials and pilot plant post audits confirmed improvement in butylene yield, gasoline octane, and coke selectivity
- Fourtiva™ delivered 1.40/bbl over previous butylene catalyst technology in refinery trials